

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8387**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	305	Seat No:	2022700034
Course Section:	1				

L_41_646_986_L1505_41470



Name of the candidate. Aditya Nair

Candidate's UID number:

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Examination class room number:

3	0	5
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Invigilator's signature with date:


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(To be filled in by the candidate)

EXAMINATION: BE - BTECH
(For example FY MCA/FY BTECH)

PROGRAM: CSE - DS

DAY AND DATE: Wednesday, October 8th I / II / III / IV / V / VI / VII / VIII

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1a

~~Q.1a~~

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

$$\begin{aligned} f'(w) &= \frac{1}{2} \times 2 \|Xw - y\| \times X \\ &= (Xw - y)X \end{aligned}$$

By gradient descent law

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial (\text{loss})}{\partial w_{\text{old}}}$$

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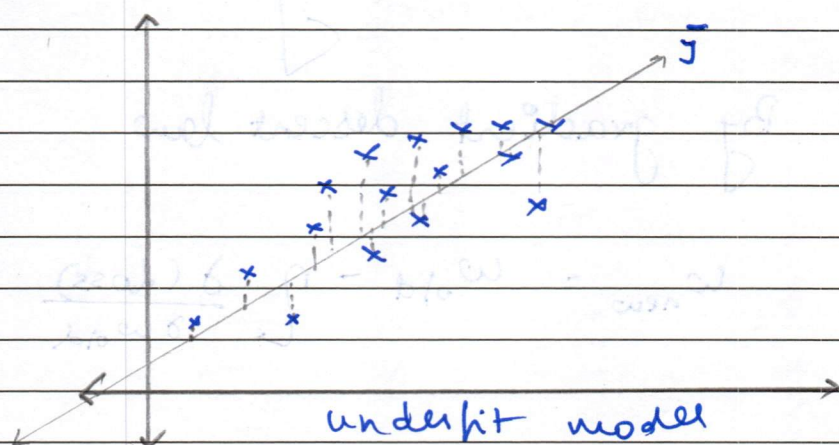
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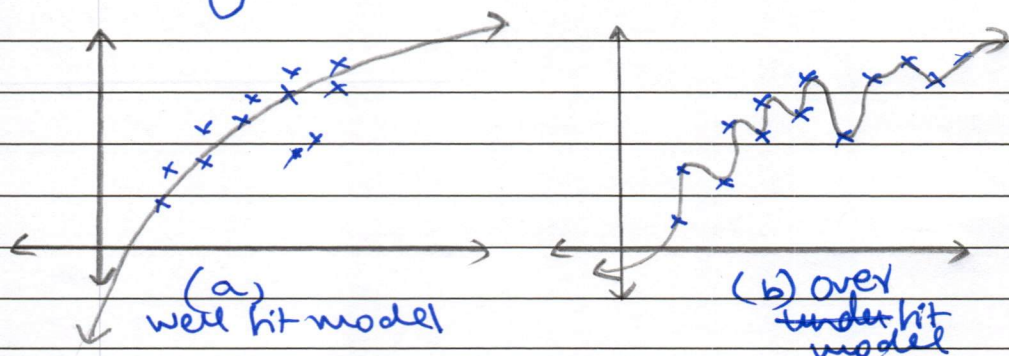
Q-16

for our predictor, let's assume
the actual value is y while
predicted value is $\hat{y}$.

$$\therefore \text{Squared error} = \sum_{i=0}^n (y_i - \hat{y}_i)^2$$



so when the model is too simple,
eg] Linear Regression, the error
coefficient is too high and
we require a model with
higher capacity to better fit our data.



here we see 2 higher capacity
models on the same data.

while (b) gives even lesser error
than (a), it has memorised
even the noise in the training data.

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→ We notice that the variance of the (b) model is very high while its bias is nearly 0.

So while it's squared error would be 0 due to it's high capacity, it's variance is nearly random and is therefore of no use for prediction.

Model (a) on the other hand, demonstrates a good balance of bias vs variance and is a good base for a prediction/generalisation model.

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Q.2a In standard 2D convolution:

~~no. of multiplication = (~~

$$\text{Height}_{op} = \frac{(H - F + 2(p))}{s} + 1$$

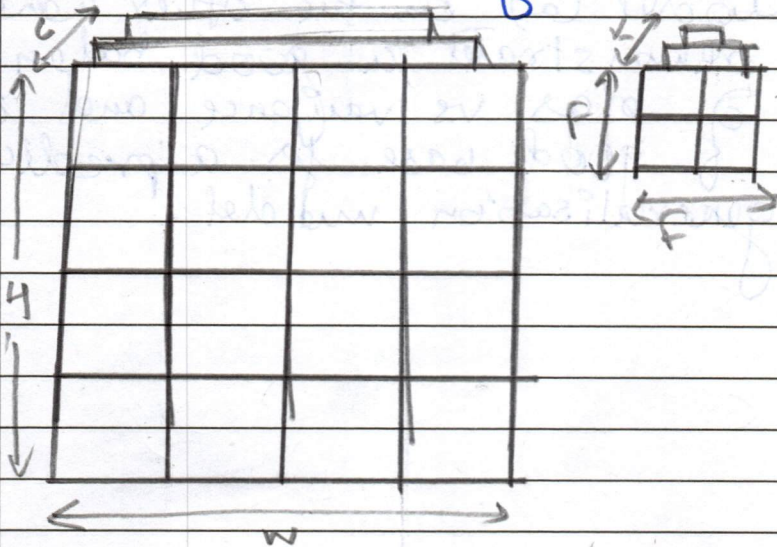
o (assume)

s (assume)

$$= H - F + 1$$

$$\text{Width}_{op} = W - F + 1$$

∴ Out put is of dimension = $(H - F + 1) \times (W - F + 1) \times K$



$$\text{no. of multiplication} = (F \times F \times C + 1) \times K$$

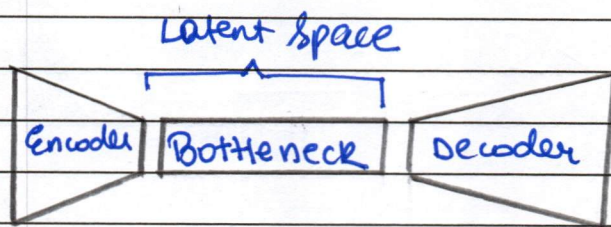
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Q.2b



In an auto encoder (AE) data is compressed down to the ~~new~~ bottleneck capacity. This is then passed through the bottleneck & decoded/reconstructed at the decoder.

now, in the case of undercomplete encoder, the number of parameters of the original data is more than the number of neurons in the bottleneck.

Thus the encoder reduces the data to such an extent that it can be represented by the lesser dimensions of the latent space.

eg] say we have 3-D data entering a 2 neuron latent space. The AE, with the help of clustering, reduces it to its 2 key dimensions & passes it through the latent space.

When reconstructed, this data is 2-D and we have successfully achieved dimensionality reduction.

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while PCA achieves it by orthogonal linear algebra, AE achieves it by clustering over epochs.

The end result, however, is similar.

Now, in the case of overcomplete AE, the no. of neurons in bottleneck $\geq$ no. of parameters in data.

So in this case, the latent space doesn't learn any new info, rather it just copies the i/p data onto its neurons & hence acting as the identity function and simply passing along the data with 0 compression.

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Q3a. The issue faced in RNN is that the older data in the RNN get multiplied again & again by its weight, and if $w < 1$, after a few rounds its value is nearly zero, i.e., no weight of old data.

How LSTM solves this is by implementing forget gate. What the forget gate does is that when new data enters the RNN, it checks if it belongs to the context. If yes it is remembered. If not it is forgotten. It also understands that when context changes and accordingly forgets old data in preference for new.

eg] "I eat a samosa every day and hence, my favorite cuisine is"

The auto complete should say "Indian" for this the key word is "samosa"

In a regular RNN however, by the time we reach "cuisine is" the weightage of "samosa" is diminished to "0".

In LSTM, "samosa" is identified as the context noun and all other words are treated as non important. Thus when we reach "cuisine is ..." we get suggested "Indian".



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Q3b. A GRU has 2 cells (update & forget) as compared to 3 cells in LSTM.

When an incoming word is detected as important, it is fed to the update cell where, o/p is given by

$$h_t = f(W_x X_t, W_{h_{t-1}} h_{t-1}) + b_t$$

where X_t = input @ t
 W_x = input weight

h_{t-1} = input from prev neuron
 $W_{h_{t-1}}$ = weight from hidden

b_t = bias.

∴ It is a convex combination of the previous hidden state & candidate hidden state.

LSTM's are much more complex than GRU's because they have memory cell, forget gate, i/p gate, o/p gate.

due to such higher no. of parameters, while highly accurate, they are slow.

GRU generalizes the concepts of an LSTM & simplifies it to just the update cell & forget cell and hence, gives comparable results while being much faster than LSTM.

Real world application: generation of sentences, autocomplete etc.

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